# Project

# ICT607 Artificial Intelligence for Cyber Security

This is a group project which has to be completed by each group that will be formed before project proposal submission. Then one of the group members can submit the project proposal.

This assessment is worth 40% of the total unit assessments. [10 marks in proposal and 30 marks in report and presentation]

Unit learning outcomes associated with this assessment:

- UNLO4: apply machine learning algorithm for a given cybersecurity task
- UNLO5: present a potential cybersecurity solution clearly in oral and written form.

**Due Date**: As stated in the LMS

**Submission**: At least one student from each group will be required to submit in the LMS via the provided link. **Note**: There are two submissions (1) the proposal Submission and (2) final report and presentation submission

**Late Penalty**: 10% per day penalty for delayed submissions unless prior extension of deadline is obtained from the unit coordinator. Work submitted more than 10 days late will not be marked. This is to enable students to receive feedback within a reasonable amount of time.

## Description of the assessment:

In this project, students are required to complete a cybersecurity task using various ML techniques. At the end, they will be required to analyse and choose the best ML solution for the task. Some of the major steps the project should include:

1. **Proposal submission:** Maximum four students can form a group. Get the approval from the teaching staff if the number of your group members is different.
2. **Cybersecurity task selection**: Depending on the available dataset you may choose any cybersecurity task (e.g. anomaly detection, intrusion detection, malware detection, threat detection etc.). This task has to be discussed with the teaching staff and clearly stated in the proposal.
3. **Dataset selection**: Students can select any dataset from their own preference but that has to be discussed and approved from the teaching staff. They also can choose any datasets from the list below. Before using a dataset, students are advised to comply with its license agreements.

- BETH: https://www.kaggle.com/datasets/katehighnam/beth-dataset
- MODBUS: https://www.kaggle.com/datasets/katehighnam/beth-dataset
- UNSW_NB15: https://www.kaggle.com/datasets/mrwellsdavid/unsw-nb15
- RT Spoofing Attacks: https://data.mendeley.com/datasets/jvgdrmjvs3/3
- 1.55M API Import for Malware analysis: https://ieee-dataport.org/open-access/155m-api-import-dataset-malware-analysis

- Fast & Furious: https://www.kaggle.com/datasets/fabriciojoc/fast-furious-malware-data-stream
- Malware API Call: https://ieee-dataport.org/open-access/malware-api-call-dataset
- Android Malware: https://www.kaggle.com/datasets/shashwatwork/android-malware-dataset-for-machine-learning
- VirtusShare: http://www.secrepo.com/Datasets%20Description/PE_malware/VirusShare.html

- SecureAge Malware: https://ncl.sg/data/public/1

4. **Pre-processing**: Acquired dataset might not be ready to be used in a ML algorithm. Make sure you perform necessary pre-processing on the dataset.
5. **Comparative study of ML techniques**: Apply at least three machine learning algorithms (or three variants of an algorithm) on the chosen dataset to perform the cybersecurity task you have selected in item 2. Perform comparative study to determine which algorithm performs the best.
6. **Accuracy**: Discuss how was the accuracy of the algorithms in your analysis
7. **Visualization and conclusion**: Finally present your findings from your analysis.

## Deliverables and rubrics:

- **Project proposal [100 marks]:** Please note, this part has a different submission deadline. This part shares 10% marks for the unit total. A typical length of a proposal can be 1500-2000 words. The proposal should include at least following components/sections:
    - **The cover page:** the first page of the report should contain the assignment names and IDs of the group members etc.
    - **Executive summary** [20 marks]: Briefly explain what you are going to do in the project.
    - **Cyber security task** [20 marks]**:** Describe with justification which task you are going to perform.
    - **Dataset** [20 marks]**:** Which dataset are you going to use in your analysis? Justify your decision.
    - **ML algorithms** [20 marks]**:** Which ML algorithms are you going to use in your analysis? Justify your decision.
    - **Potential challenges** [20 marks]**:** Describe some potential challenges to complete the task.
- **Final report [75 marks]:** Complete the final report with at least the following sections:
    - **The cover page:** the first page of the report should contain the assignment names and IDs of the group members etc.
    - **Executive summary [10 marks]**: Briefly describe what you have done in this project and what were your findings.
    - **Dataset selection and pre-processing** [15 marks]
    - **Comparative study of ML techniques** [20 marks]
    - **Visualization and conclusion** [15 marks]: Present your findings from your analysis.
    - **Code explanation** [15 marks]: Briefly explain how your code should be executed. This should include a brief explanation of all necessary files and dependencies, location of the dataset, necessary library installations etc.
- **Code [5 marks]:**
    - Submit the code and necessary dependency files. Make sure you organize your code in such a way that the marker can execute your code smoothly.

- **Final presentation [20 marks]:** Each group will get around 20 minutes to present their work using PowerPoint slides. This presentation will take place in the last lab class of the semester. The report has to be submitted before the presentation.

**Submission**: Submission in two stages:

**Stage 1:** Submit the proposal document

**Stage 2**: Zip the final report, code and presentation in a zip folder and submit it in the LMS.

**Note:** For any further clarification or feedback please feel free to contact the teaching staff.